

Joey Dufourd, Ph.D.

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Research interests: *Life Science / Computational Biology & BioInformatics / Chromatin & Telomeres / Multi-omics / Machine Learning / Computer Vision / Big data & Data Science*

Profile

Molecular biologist with computational expertise, passionate about understanding life through science. My PhD on telomere biology combined advanced molecular and imaging approaches (3D-SIM, proteomics) with custom computational pipelines in Python and R for image analysis, statistical modeling, computer vision and data integration. Proficient in Bash, Python, and R, I leverage computational approaches to develop reproducible analytical workflows and explore complex biological systems. I am convinced that integrating the strengths of wet-lab experimentation with the power of computational approaches will accelerate discoveries and enable transformative advances in biomedical research.

Research Experience

PhD Student — Repeated sequences chromatin & Computer vision 10/2021–12/2025
Institut de Génétique Humaine (CNRS), Montpellier, France **Supervisors:** J. Déjardin, M. Tardat

- **Discovered and characterized TELS1**, a telomere binding & mouse embryonic stem cell specific protein that stabilizes *t-loops* independently of TRF2.
 - **Combined** PICCh/IP-MS, ChIP, FISH (nuclei & metaphases), microinjection of mouse embryos, CRISPR-Cas9, 3D-SIM super-resolution microscopy and other approaches to investigate the role of TELS1 at telomeres.
 - Built **automation & AI pipelines** (computer vision) to automate *t-loop* quantification and microscopy analysis (CellProfiler).
 - Integrated imaging, proteomics, statistics and molecular data using **Python & R**.
 - **Conducted statistical modeling and experimental design** (ANOVA and multiple-testing aware post-hoc).
 - **Co-founded and currently coordinate** the IGH Young Scientist Club (YSC), a monthly forum promoting collaboration and knowledge sharing among early-career researchers.
 - Delivered **seminars and workshops** on computational biology, statistics, and artificial intelligence for students and researchers at IGH.
 - **Taught** undergraduate courses in genetics and epigenetics.
- **Tech stack:** Python (TensorFlow/PyTorch, scikit-learn, pandas, NumPy, Matplotlib,...), R (tidyverse, ggplot2, Bioconductor, ...), Bash, 3D-SIM image processing, AlphaFold modeling, Git, statistical modeling (ANOVA, clustering, PCA), CellProfiler, ImageJ/Fiji.

M2 Internship — Repeated Sequences Chromatin 02/2021–08/2021
Institut de Génétique Humaine (CNRS), Montpellier, France

- Investigated telomeric chromatin structure and *t-loop* stability in **mouse embryonic stem cell and embryo-like structures**.
- **Analyzed sequencing data**, performed R-based statistical analyses, and quantified images using CellProfiler (normality tests, ANOVA, Bonferroni-corrected tests).
- **Integrated experimental and computational approaches** to identify novel regulators of telomere structure and chromatin dynamics.

M1 Internship — Stanford University (Van Rechem Lab)
Stanford Medicine, USA

02/2020–08/2020

- Characterized non-canonical cytoplasmic functions of **PRMT5** in cancer biology; applied RNAi, polysome profiling (sucrose gradients), and confocal imaging.
- Managed and performed experiments across >10 different cancer cell lines in parallel, **ensuring reproducibility and consistency of results**.

iGEM Team Leader & Project Inventor — Montpellier

11/2018–11/2019

International Genetically Engineered Machine (iGEM) Competition — Gold Medal

Project: KARMA — an innovative molecular system for targeted protein degradation.

- Conceived and led the **KARMA project**, a modular tool combining specific recognition antibodies with a targeted degradation protease to selectively remove proteins, with potential applications in cancer and antibiotic resistance.
- **Supervised and coordinated** a multidisciplinary team of 14 students (biology, bioinformatics, and engineering) through project design, cloning, modeling, and validation phases.
- Built and maintained the **project website and wiki**, ensuring clear and accessible communication of objectives and results (igem.org/Team:Montpellier).
- **Led public engagement initiatives** (*Carascience*) to present the project and basic biology to schools and the general public.
- Strengthened skills in **project management, communication, synthetic biology, bioinformatic coding, statistical analysis, and web development** within an international competition framework.

Research Intern — CNRS (DIMNP & IGF, Montpellier)

10/2018–12/2018

Supervised by Prof. Ovidiu Radulescu (DIMNP) — Project on mathematical modeling of HIV latency and apoptosis mechanisms.

- Designed and analyzed **multiscale computational models** of HIV latency to explore viral persistence and reactivation dynamics across molecular, cellular, and population scales.
- Applied **stochastic simulations (Gillespie algorithm)** and differential modeling to investigate latency transitions; developed reusable scripts later adapted to model telomere state dynamics (github.com/joeydu).
- Strengthened skills in **quantitative biology, dynamical systems, and data analysis** applied to virology and cell fate modeling.

Publications & Manuscripts

- 📄 **Dufourd J.** et al. *TELS1 stabilizes telomeric t-loops independently of TRF2 in pluripotent cells.* (2025).
- 📄 **Dufourd J.** *AI-assisted image analysis for telomeric structure detection and quantification.* In preparation (2025?).

Projects Highlights

- **AI-assisted image analysis for telomere structure** — Developed a YOLO-based computer vision pipeline for *t-loop* & line detection (2025).
 - **Stochastic simulations (Gillespie algorithm)** — Stochastic simulation of telomere state transitions using the Gillespie SSA; explores dynamics of *t-loop* formation/processing and end-protection; includes a reproducible script for parameter sweeps and seeded runs. ( [joeydufourd/telomere-dynamics-gillespie](#))
 - **FIJI-Macros (ImageJ)** — Open-source macro suite to automate multichannel microscopy workflows (batch conversion, channel alignment, montage generation,...). ( [joeydufourd/FIJI-Macros](#))
 - **Discovery of TELS1** — Identified a pluripotency-specific *t-loop* stabilizing protein through unbiased & high-throughput PICCh experiment (PhD).
 - **KARMA (iGEM 2019)** — Led the development of a modular protein degradation system; awarded Gold Medal.
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Education

PhD in Genetics, Epigenetics and Cellular Determinism Université de Montpellier — Institut de Génétique Humaine (CNRS), France	2021–2025
MSc in Genetics & Cellular Biology Université de Montpellier, France	2019–2021
BSc in Molecular Biology Université de Montpellier, France	2016–2019

Skills

- **Computational:** Python (NumPy, pandas, TensorFlow, PyTorch), R (tidyverse, ggplot2, Bioconductor), Bash, Image analysis (CellProfiler, Fiji), Computer vision, Machine & Deep Learning, Statistics,
 - **Experimental:** CRISPR/Cas9, ChIP, PICCh, FISH (nuclei/metaphase), 3D-SIM microscopy, Proteomics, RT-qPCR, Immunofluorescence,
 - **Data Science:** Multi-omics integration, statistical modeling (ANOVA, linear models, clustering/PCA), reproducible pipelines, documentation,
 - **Communication:** Teaching (genetics, epigenetics), oral presentations, mentoring, scientific writing.
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Teaching

-  Undergraduate teaching (Genetics & Epigenetics): lectures and practicals for 2nd–3rd year students.
-  Workshops and seminars on computational biology: taught introductory sessions on programming (Python, R), statistics, machine learning, and artificial intelligence for life scientists.